

# Agentic Delegation

and the Language Frontier

Quispe & Xu (2026), arXiv:2605.25438v2

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## Question

Does agentic AI expand the set of programming languages in which a developer can ship code, rather than only accelerating familiar work?

**Citation audit.** The current paper has two authors, Alexander Quispe and Kevin Xu, and a new title. It is a preliminary, unrefereed [arXiv preprint](#), not an NBER paper. **Claim boundary.**

This is a [production frontier](#), not proof that the developer learns an unfamiliar language.

# Three capabilities, one threshold mechanism

## Solo $S$

Human executes with language-specific skill  $s$ .

## Augment $C$

Suggestions help in proportion to the foothold  $s$ .

## Delegate $D$

Agent executes; human specifies and verifies using general ability  $a$ .

$$M^1 = \{S, C\} \subset M^2 = \{S, C, D\}, \quad Z^g = 1 \left\{ \max_{m \in M^g} V^m \geq 0 \right\}.$$

The core model has discrete mode choice and entry—**no continuous effort choice**.

# Main result: an activation band

For an unfamiliar language, augmentation needs a foothold:

$$\gamma s - r_C \leq 0 \implies T^1 = T^S.$$

If delegation lowers the threshold,  $B = T^S - T^D > 0$ , then

$$Z^2 - Z^1 = 1\{T^D \leq \omega < T^1\}.$$



## Estimation panel

**5,346** developers: 2,813 treated and 2,533 not-yet-treated controls; 28 months and 149,688 rows.

**Measurement:** 3.2m commits, 133k developer–repository pairs, and **57m changed files** classified with GitHub Linguist.

Adoption is first detectable Claude co-authorship. Estimation uses doubly robust staggered DiD, one-month anticipation, and developer-clustered bootstrap SEs.

## Identification

Voluntary adoption may coincide with a new unfamiliar-language project. Estimates are associations, not definitive causal effects.

# What happens at adoption?

| Outcome              | ATT at $e = 0$ | SE    | Benchmark        |
|----------------------|----------------|-------|------------------|
| Active languages     | <b>2.528</b>   | .063  | pre-mean .90     |
| Newly used languages | <b>1.193</b>   | .051  | pre-flow .31     |
| Language entropy     | <b>.382</b>    | .009  | pre-mean .15     |
| Cumulative languages | <b>1.604</b>   | .054  | 2.072 by $e = 2$ |
| Monthly commits      | 35.080         | 2.085 | diagnostic       |

## Stock–flow signature

New-language entry spikes and reverts; cumulative language breadth continues growing. Specialists add far more new languages than generalists within activity groups.

Agentic delegation can turn **general specification-and-verification ability** into production across domains where specialized execution skill is scarce.

Robustness checks and specialist heterogeneity support the mechanism. They do not eliminate selection into adoption or prove skill acquisition.

Full derivations, SymPy audit, and extended deck:  
<https://github.com/johnbarraza/ai-03-quispe>